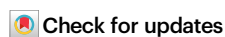


Assimilative causal inference

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Causal inference is fundamental across scientific disciplines, yet existing methods struggle to capture instantaneous, time-evolving causal relationships in complex, high-dimensional systems. In this paper, assimilative causal inference (ACI) is developed, which is a methodological framework that leverages Bayesian data assimilation to trace causes backward from observed effects. ACI solves the inverse problem rather than quantifying forward influence. It uniquely identifies dynamic causal interactions without requiring observations of candidate causes, accommodates short datasets, and, in principle, can be implemented in high-dimensional settings by employing efficient data assimilation algorithms. Crucially, it provides online tracking of causal roles that may reverse intermittently and facilitates a mathematically rigorous criterion for the causal influence range, revealing how far effects propagate. The effectiveness of ACI is demonstrated by complex dynamical systems showcasing intermittency and extreme events. ACI opens valuable pathways for studying complex systems, where transient causal structures are critical.

Causal inference determines cause-and-effect relationships between variables^{1–3}. It has found wide applications across different disciplines such as atmospheric and ocean science, economics, and neuroscience^{4–6}. In addition to discovering the interactions between variables, causal inference plays a significant role in model identification, policy evaluation, and decision-making^{7–11}.

Methods for causal inference can be classified into different categories depending on the available resources for use. On the one hand, a natural way to identify the overall causal structure is to exploit multivariate time series, where temporal dependencies are often utilized to infer causal relationships. Methods falling into this category include Granger causality¹², transfer entropy^{13,14}, mutual information^{15,16}, convergent cross mapping^{17,18}, causation entropy^{19,20}, and mixture causal discovery²¹. On the other hand, models built upon physics can assist in understanding causal dependence between different variables. Information transfer based on the ensemble forecast of the underlying model has been exploited to indicate certain causal relationships over short terms²². In addition, linear response theory, which infers causal links by analyzing system responses to small perturbations at the equilibrium, has been utilized to reveal the attributions of variations in climate systems^{23–25}. Other recently developed methods include using Koopman operators for causal discovery²⁶,

causal graphs²⁷, dynamic neural relational inference²⁸, and approaches based on machine learning^{29–31}.

Nature can be regarded as a complex dynamical system where only a single random realization from this system is available. Due to the underlying strong nonlinearity and multiscale features, such a realization is usually intermittent and stochastic. Detecting instantaneous causal relationships as a function of time is then crucial for understanding the dynamic interplay among variables, where the roles of causes and effects can shift repeatedly over time and at irregular intervals. In environmental science, extreme events, such as the triggering mechanisms of hurricanes and the abrupt transitions in weather patterns, have significant scientific and social impacts. Understanding the precursors of each extreme event and its subsequent implications requires online tracking of the causal relationship between them. Likewise, instantaneous causal analysis is of broad interest in neuroscience to discover temporary role reversals between brain regions during decision-making or cognitive tasks. However, most purely data-driven causal inference methods exploit data points over a long time series to reveal the average causal directions. In contrast, purely model-based methods, such as computing the information transfer in²², utilize ensemble forecasts to infer the causal relationship when the system begins from a given initial condition and evolves toward

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statistical equilibrium. Yet, these methods are typically only able to provide time-dependent results for a very short time and become computationally challenging as the dimension of the system increases. Additionally, these model-based methods often struggle to optimally handle cases with a single observed realization.

Given the importance of discovering instantaneous causal relationships, this paper introduces a methodological framework called assimilative causal inference (ACI) to compute time-dependent causal interactions in complex systems. As in many practical applications, only a single realization of a subset of the state variables is available as observations. Meanwhile, an associated model, often turbulent and stochastic, is accessible as a supplement. Fundamentally different from traditional causal inference methods, which treat causality as a forward problem (analyzing how causes propagate into effects), ACI instead addresses it as an inverse problem. Using Bayesian data assimilation, ACI traces causes backward from observed effects by quantifying whether incorporating information about the effects reduces uncertainty in reconstructing the potential causes.

ACI has several desirable advantages. First, it captures the evolving interplay among state variables, as their causal and effect roles can be reversed across time. Thus, ACI provides direct insights beyond time-averaged causal links, making it particularly valuable for studying turbulent systems with intermittency and regime switching. Second, ACI identifies the causal inference range (CIR) of a relationship, revealing how far causal effects propagate at each time instant. The framework includes a mathematically justified criterion to determine the CIR without empirical thresholds. While the importance of the CIR has been highlighted in complex networks^{32,33}, ACI offers a unique and general method for its computation in turbulent systems with varying auto- and cross-correlation decay rates. Third, ACI scales efficiently to high-dimensional problems by leveraging computational methods developed in Bayesian data assimilation. This is fundamentally different from information-transfer methods limited to low dimensions²². In addition, ACI can accommodate short time series and incomplete datasets, which often appear in geophysics and climate science. Notably, ACI does not require observations of candidate causes, which is a key advantage since potential drivers are usually unknown or unmeasured. Instead, uncertainty reduction in potential causal variables can be determined solely from the observed effects and governing dynamical model via Bayesian data assimilation. As a final remark, while data assimilation has been used to assist in detecting the attribution of weather and climate-related events^{34,35}, these studies primarily focus on estimating the state in response to specific external perturbations, rather than online causal inference or CIR estimation. ACI also fundamentally differs from these methods in its integration of data assimilation into causal inference.

The remainder of this paper is organized as follows. Numerical results from applying ACI on various nonlinear complex dynamical systems with intermittency and extreme events are presented in Section 2 (Results). Discussions and remarks on the developed methodology are contained in Section 3 (Discussion). The operational details of the ACI framework are included in Section 4 (Methods): Section 4.1 presents the mathematics of the ACI framework, followed by the discussion of CIR in Section 4.2. Section 4.3 extends the ACI framework to account for the presence of non-target variables. Mathematical derivations and additional test results are available in the Supplementary Information.

Results

We apply the ACI framework to identify causal relationships across the state space and evaluate the corresponding CIRs in complex nonlinear dynamical systems exhibiting intermittency and extreme events. The dynamical properties of these systems and the mechanisms that trigger crucial observed phenomena are well understood. This knowledge is exploited as guidance to validate the results when applying ACI. In

addition, a perfect-model setup is used in the case studies carried out, meaning the system generates the synthetic observational data and the forecast model in ACI at the same. This facilitates the recovery of causal relationships in the absence of model error. Because the underlying system is chaotic and subject to stochastic forcing, a perfect model means that the model used in ACI shares the same governing equations and long-term statistical behavior as the system that generates the true signal. Additional results, including further numerical experiments, case studies, and one crucial application using sparse real-world observations, are provided in the Supplementary Information.

The operational outline of the ACI metric used to infer instantaneous causal relationships in the following experiments, including the details of how we objectively determine the associated CIRs and account for non-target variables, is described in Section 4 (Methods). The developed framework is also summarized via the schematic diagram provided in Fig. 3. Additional mathematical results and analyses, as well as computational implementation details, can be found in the Supplementary Information.

A nonlinear dyad model with intermittent extreme events

Let us start with a two-dimensional model, which nevertheless has strong nonlinear features with observed extreme events. The model reads as:

$$\frac{dx}{dt} = -d_x x + \gamma xy + f_x + \sigma_x \dot{W}_x, \quad (1a)$$

$$\frac{dy}{dt} = -d_y y - \gamma x^2 + f_y + \sigma_y \dot{W}_y, \quad (1b)$$

where $\dot{W}_x$ and $\dot{W}_y$ are independent (in time and mutually) standardized normal random variables. Within the proposed ACI framework, only x is assumed to be observed over time. This is a reduced-order conceptual model for atmospheric variability. It has been used to analyze the effects of various coarse-grained procedures on processes exhibiting intermittency, large-scale bifurcations, and microscale phase transitions. It is defined by an energy-conserving condition on its quadratic nonlinearities³⁶. The following parameter values are used for this model:

$$\begin{aligned} d_x &= 0.5, \gamma = 2, f_x = 0.5, \sigma_x = 0.5, \\ d_y &= 0.5, f_y = 1, \sigma_y = 1. \end{aligned} \quad (2)$$

Figure 1 illustrates the ACI and CIRs from y to x . Panel (a) shows that extreme events in x occur intermittently. When the combined coefficient $-d_x + \gamma y$ in (1a) becomes positive, the dynamics of x exhibit anti-damping, leading to amplitude growth and extreme events. Conversely, when the coefficient is negative, x behaves as a damped process, with fluctuations primarily driven by random noise. Panel (b) displays the ACI, revealing that phases with significant ACI values largely coincide with the onset and peak phases of extreme events in x . This aligns with intuition, as y acts as an anti-damping source and the primary driver of these events. Once x peaks and begins to decay, the ACI value drops sharply, reflecting the strong negative feedback from x to y , which then dampens x . The whisker plot in Panel (a) highlights the objective CIR, indicating a sustained influence from y to x during the triggering phase of extreme events but minimal influence during their demise. Notably, the long-range influence from the CIR reveals that extreme events develop gradually, with triggering conditions established well in advance. From a data assimilation perspective, when the observable signal x starts to strengthen, it enhances the ability of the smoother to estimate y as the signal in the future contains useful information beyond noise. As a result, strong ACI and longer CIR values are obtained. In contrast, once x becomes sufficiently strong, its

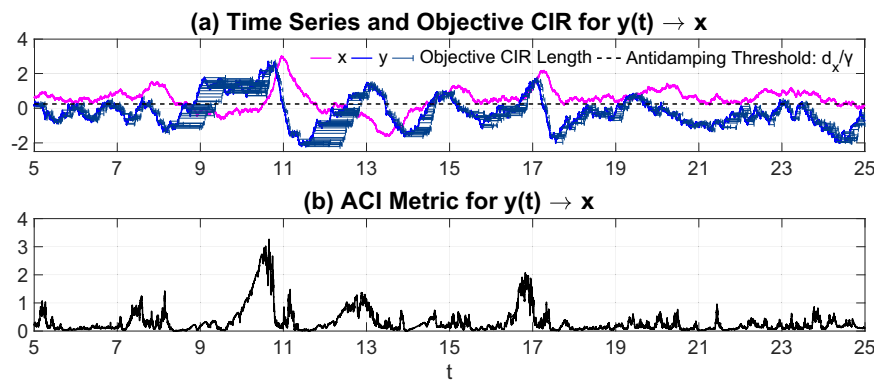


Fig. 1 | ACI values and CIRs from y to x at each time instant t for the nonlinear dyad model (1). **a Time series of x (magenta) and y (blue), the objective CIR represented by a whisker plot emanating forward in time from each $y(t)$ value, and the anti-damping threshold line at d_x/y . **b** ACI from y to x as a function of time.**

high signal-to-noise ratio ensures the filter captures all relevant triggering dynamics, leaving little room for the smoother to improve estimates. Hence, the outset of the ACI metric's decline and shorter CIRs. The Supplementary Information contains the filter and smoother distributions and the subjective CIRs with different threshold values ε .

A stochastic model capturing the El Niño–Southern Oscillation (ENSO) diversity

El Niño–Southern Oscillation (ENSO) is a dominant climate phenomenon characterized by quasi-regular periodic behaviors in sea surface temperatures (SSTs) and in the atmospheric circulation across the equatorial Pacific Ocean^{37,38}. ENSO exhibits remarkable diversity in its spatial patterns, temporal evolution, and impacts, which can be broadly categorized into two main types: Eastern Pacific (EP) and Central Pacific (CP) El Niños, where an anomalous warming center occurs in the eastern and central Pacific, respectively³⁹. The opposite phases with anomalous cooling SSTs are called La Niña. Understanding the causes of different ENSO events is crucial for improving climate predictions and mitigating socio-economic consequences.

Although few models can accurately capture ENSO diversity, a recently developed stochastic conceptual model successfully reproduces its diverse behaviors and non-Gaussian statistics⁴⁰. This model has been highlighted in a recent review⁴¹, making it a suitable testbed for studying the ENSO diversity. Mathematical details are provided in the Supplementary Information. The model consists of six state variables: ocean zonal current in the CP (u), western Pacific (WP) thermocline depth (h_W), CP SST (T_C), EP SST (T_E), atmospheric winds (τ , intraseasonal), and background Walker circulation (I , decadal). The first four variables (u , h_W , T_C , and T_E) operate on interannual time-scales. As a nonlinear system with state-dependent noise, the model generates extreme events and intermittency. The two SST variables allow reconstruction of spatiotemporal SST patterns across the equatorial Pacific, providing an intuitive way to identify different ENSO event types.

The variables T_C , h_W , and τ directly influence T_E . Figure 2 displays their conditional ACI and CIR values to T_E (non-target variables have been resolved via the framework proposed in Section 4.3). For EP El Niño events (positive T_E anomalies, shown in red), T_C exhibits the strongest ACI value, slightly preceding the T_E peak. This timing aligns with physical understanding, as SSTs in these regions are strongly coupled: during El Niño, warm water propagates from CP to EP, producing the observed ACI lead. The τ to T_E ACI value appears noisier due to τ 's intraseasonal variability, yet its evolution confirms τ 's robust impact on T_E . Winds drive warm water propagation and exert near-instantaneous SST effects. In contrast, while h_W significantly contributes to T_E , its ACI amplitude is weaker than T_C or τ . The discharge–recharge theory⁴² posits h_W – T_E oscillator dynamics, but in

models with refined CP variables, h_W 's influence on T_E becomes indirect: h_W first affects T_C , which then propagates signals to T_E (WP $\rightarrow$ CP $\rightarrow$ EP). Consistent with this paradigm, h_W 's ACI value to T_E peaks months before EP El Niño maxima. CIRs further corroborate these physical mechanisms: T_C shows the longest influence, h_W 's more indirect role yields intermediate CIRs, and τ 's intraseasonal nature produces the shortest impacts. The Supplementary Information provides additional ACI and CIR analyses for all variables across different ENSO event types. In addition, in the Supplementary Information, a case study using real-world ENSO observations is examined, which does not assume a perfect model setup. The conclusions derived about the causal relationships using real-world observational data remain qualitatively similar to those with model-generated data. The results provide preliminary evidence that the ACI framework remains reasonably robust under realistic conditions, including observational noise, moderate model error, and numerical discretization effects.

Discussion

In this paper, the assimilative causal inference (ACI) framework is developed for the detection of instantaneous causal relationships and of their associated causal influence ranges (CIRs) in complex dynamical systems. ACI leverages Bayesian data assimilation to trace causes backward from observed effects. It uniquely identifies dynamic causal interactions without requiring observations from candidate causes, accommodates short datasets, and, in principle, can be scaled efficiently to high dimensions using efficient Bayesian data assimilation algorithms. Crucially, it provides online tracking of causal roles, which may reverse intermittently, and facilitates a mathematically rigorous criterion for the causal influence range, revealing how far causal impacts propagate temporally. Its ability to dynamically detect causal influence ranges and handle high-dimensional systems by employing computationally efficient Bayesian techniques highlights its potential for applications in climate science, neuroscience, and other fields. Numerical tests on nonlinear systems with extreme events and regime transitions demonstrate the effectiveness and robustness of the framework.

It is worth highlighting how ACI differs from other state-of-the-art methods in terms of expressivity, computational complexity, and underlying assumptions. Different from most methods that identify cause-and-effect relationships by running the system forward and quantifying informational responses in the target variables' entropy conditioned on past candidate causes, which is a predictive setting typical of information-flow-based methods, ACI instead formulates causality as an inverse problem in uncertainty quantification. Within the Bayesian data assimilation framework, candidate causal variables are inferred backward from observations of their potential effects. Thus, while many existing methods forecast causes forward in time,

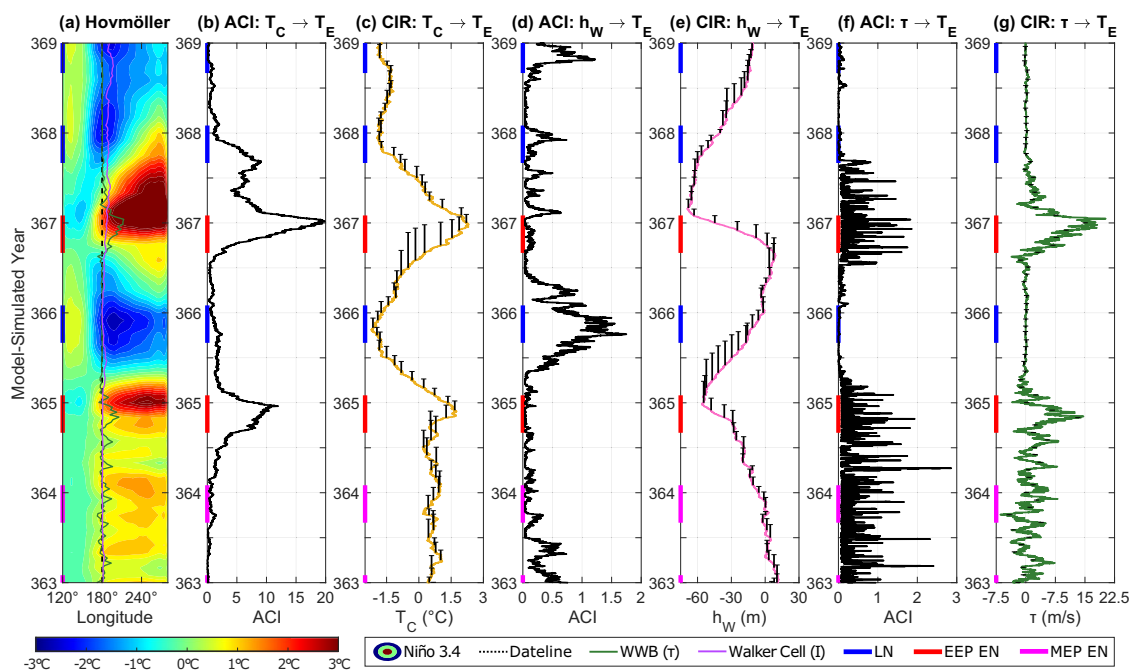


Fig. 2 | Conditional ACI values and CIRs of the stochastic conceptual model for the ENSO diversity with T_E as the effect variable over a six-year model-simulated period (where EEP EN and MEP EN stand for extreme and moderate EP El Niños, respectively, and LN for La Niña). a Hovmöller diagram of the SST anomalies in the equatorial Pacific reconstructed from T_C and T_E via a

spatiotemporal multivariate regression technique⁴⁰. **b** ACI values from T_C to T_E as a function of time. **c** Time series of the observable T_C and the objective CIR represented by a whisker plot emanating forward in time from the value of T_C at each month over this six-year period. **d, e** Same as **(b)** and **(c)** but for the causal link from h_W to T_E . **f, g**: Same as **(b)** and **(c)** but for the causal link from τ to T_E .

ACI instead traces causes back from observed effects, effectively interpolating rather than extrapolating causality. This distinction makes ACI more expressive in capturing instantaneous causal structures and, in practice, often more robust. From a computational standpoint, ACI can leverage well-developed Bayesian data assimilation tools that scale effectively to high-dimensional systems^{43,44}. In contrast, information transfer and other model-based methods often require computing the exact density functions and their derivatives, which is computationally demanding and subject to the curse of dimensionality. Finally, ACI requires only a single time series but assumes access to a dynamical model, which enables it to identify instantaneous causal relationships. In contrast, most time-series-based methods operate under model-free assumptions and therefore focus on aggregated or time-averaged causal links rather than on causality at specific moments. Because ACI incorporates model dynamics, it does not require long time series to detect causes and effects. Instead, it can infer instantaneous causal links from a single data point combined with model information.

Future work will focus on several important directions. First, because model error is ubiquitous in practice, it is essential to study its impact on causal inference. Understanding how model error affects inferred causal relationships can create opportunities not only to detect and correct model error, but also to leverage this information to develop parsimonious models for complex dynamical systems. Similarly, observational data may contain additional noise that can affect causal inference. The ACI framework provides a natural way to analyze how observational noise contaminates causal detection. In this context, data assimilation can serve as a pre-filtering step to reduce the influence of observational noise, which allows for a systematic examination of its role in shaping causal inference outcomes. Second, by building on the computationally efficient methods developed in Bayesian data assimilation, it is essential to apply ACI to higher-dimensional systems. Ensemble data assimilation is a well-established technique that routinely uses localization, covariance inflation, and other adjustments to overcome the curse of dimensionality^{45–47}. As a

result, the ensemble size (typically on the order of $O(100)$) is often chosen independently of the system's dimension, which contributes to the scalability of Bayesian data assimilation and, consequently, of ACI. However, the sampling errors inherent in ensemble methods may introduce additional biases. Thus, it is necessary to examine whether such errors affect ACI. In addition, various coarse-grained approaches, such as approximating the full distribution with a Gaussian, are widely used in practice. Although these approximations introduce errors in computing the PDF, they are robust and highly accurate when computing diagnostic indicators, such as the relative entropy, that are used for model identification and sensitivity analysis. Thus, they may substantially accelerate ACI in practice. Third, building upon the current forward-in-time CIR analysis, the framework allows for the development of a crucial complement that focuses on backward-in-time event attribution to past causes. The forward and backward CIR tools would support distinguishing different types of tipping points in Earth systems, identifying the causes of regime switching, and detecting early warning signals. Finally, continuous-in-time observations are adopted in this work for simplicity of presentation. Nevertheless, an analogous framework can be naturally extended to discrete-in-time observations using the corresponding Bayesian data assimilation algorithms. An interesting scientific question in this context is to understand the uncertainty in causal detection arising from the discrete nature of the data, particularly at time intervals between successive observations. This line of inquiry could also highlight an important distinction between ACI and purely data-driven methods. In ACI, the dynamical model provides a natural way to interpolate causal relationships at time instants when observations are unavailable. Therefore, it may help bridge observational gaps and maintain temporal continuity in causal inference.

Methods

In this section, we provide a high-level overview of the methodology underlying the ACI framework, including its formulation of causal relationship assessment through a Bayesian inverse problem and the

evaluation of the associated CIRs via objective means. We also outline the strategy used to infer causal links while accounting for interference from non-target or ancillary variables. Rigorous mathematical justifications are provided in the Supplementary Information.

The assimilative causal inference (ACI) framework

Setup and notation. Consider a time interval $[0, T]$ and a specific time instant t , where $0 \leq t \leq T$. Let $\{\mathbf{x}(t)\}_{0 \leq t \leq T}$ and $\{\mathbf{y}(t)\}_{0 \leq t \leq T}$ be two multivariate stochastic processes defined over this interval. Denote by $\mathbf{x}(t)$ and $\mathbf{y}(t)$ the corresponding multivariate random variables at time t . Their evolution in time is defined by the following stochastic dynamical system:

$$\begin{aligned} \frac{d\mathbf{x}}{dt} &= \mathbf{f}^x(t, \mathbf{x}, \mathbf{y}) + \Sigma_1^x(t, \mathbf{x})\dot{\mathbf{W}}_1(t) + \Sigma_2^x(t, \mathbf{x})\dot{\mathbf{W}}_2(t), \\ \frac{d\mathbf{y}}{dt} &= \mathbf{f}^y(t, \mathbf{x}, \mathbf{y}) + \Sigma_1^y(t, \mathbf{x}, \mathbf{y})\dot{\mathbf{W}}_1(t) + \Sigma_2^y(t, \mathbf{x}, \mathbf{y})\dot{\mathbf{W}}_2(t), \end{aligned} \quad (3)$$

where $\dot{\mathbf{W}}_1$ and $\dot{\mathbf{W}}_2$ are independent (in time and mutually) mean-zero Gaussian random vectors with uncorrelated components that possess unit variance. Mathematical details regarding the structure of (3) are given in the Supplementary Information. For notational simplicity, the dependence on t is sometimes omitted when referring to these state variables and no distinction between random variables and their realizations is made. Let $\mathbf{x}(s \leq t)$ represent one realization of the stochastic process $\{\mathbf{x}(t)\}_{0 \leq t \leq T}$ as a time series over the interval $[0, t]$. Similarly, $\mathbf{x}(s \leq T)$ denotes a realization of $\{\mathbf{x}(t)\}_{0 \leq t \leq T}$ as a time series over the entire interval $[0, T]$. For simplicity, the time series are assumed to be continuously observed. An analogous framework can be developed for discrete-in-time observations. Throughout this paper, the time series is assumed to be free of observational noise.

According to Granger's predictive causality or its nonlinear extension, transfer entropy, $\mathbf{y}$ is the cause of $\mathbf{x}$ at time t if the knowledge of $\mathbf{y}(s \leq t)$ improves the prediction or reduces the uncertainty of $\mathbf{x}(t)$ ^{12–14}. Using information theory, this can be expressed as:

$$S(p(\mathbf{x}(t)|\mathbf{y}(s \leq t))) < S(p(\mathbf{x}(t))), \quad (4)$$

where $S(\cdot)$ is Shannon's entropy and the history of the target variable in the conditioning, i.e., $\mathbf{x}(s < t)$, is omitted in (4) for notation simplicity.

Causal discovery from a Bayesian inverse problem viewpoint. The above argument can be interpreted in a reverse way, from the perspective of statistical inference through an inverse problem in uncertainty quantification. If $\mathbf{x}$ is the subsequent effect of $\mathbf{y}(t)$, then knowing future information about $\mathbf{x}$ reduces the uncertainty in inferring the current state of $\mathbf{y}$:

$$S(p(\mathbf{y}(t)|\mathbf{x}(s > t))) < S(p(\mathbf{y}(t))). \quad (5)$$

Further, including the past and current information of $\mathbf{x}$ on both sides yields

$$S(p(\mathbf{y}(t)|\mathbf{x}(s \leq T))) < S(p(\mathbf{y}(t)|\mathbf{x}(s \leq t))), \quad (6)$$

which does not change the causal relationship in (5).

A notable feature of (6) is that it establishes a connection between causal discovery and Bayesian data assimilation. Suppose the underlying turbulent and potentially stochastic model governing $\mathbf{x}$ and $\mathbf{y}$ is known and given by (3). Running the model forward provides a statistical estimation of the state of $\mathbf{y}(t)$ (known as 'forecast'). However, this state estimation differs when also incorporating the knowledge of the time series $\mathbf{x}(s \leq t)$ or $\mathbf{x}(s \leq T)$ (known as 'analysis'), as these specific observations provide additional information to reduce the uncertainty when inferring the state of $\mathbf{y}(t)$. This two-step (forecast-analysis)

pipeline is known as Bayesian data assimilation^{44,48}, where the distribution derived solely from the model statistics is referred to as the prior, while the observed time series is used to compute the likelihood, accounting for the uncertainty in the observational process. The two conditional distributions in (6) result from combining the model-based information with the observation-induced likelihood and are referred to as the posteriors. Specifically, the distribution on the left-hand side of (6) corresponds to smoothing, while the one on the right-hand side corresponds to filtering^{49,50}.

ACI. Denote by p_t^s and p_t^f the smoother and filter posterior distributions at time t (i.e., the two distributions on the left- and right-hand sides of (6)), respectively. If the uncertainty reduction in p_t^s is more significant than that in p_t^f , then it is due to the incorporation of the future information of $\mathbf{x}$, which indicates toward the contribution from $\mathbf{y}(t)$ to the future states of $\mathbf{x}$.

Although the entropy difference has been widely used in many other causal inference methods, the relative entropy is employed hereafter to quantify the uncertainty reduction in p_t^s related to p_t^f . In addition to the covariance (and other higher-order moments), which measure the variability of the random variable, the relative entropy also accounts for the difference in the mean state of the two distributions. This difference is crucial for reflecting the additional information gained by incorporating future observations. Furthermore, it is coordinate-free, i.e., it is invariant under general nonlinear changes of the state variables. Notably, relative entropy has been widely used to assess uncertainty reduction in the context of data assimilation^{51–54}.

As such, if the relative entropy between the smoother and filter distributions is nonzero, i.e.,

$$\mathcal{P}(p_t^s, p_t^f) = \int p_t^s \ln(p_t^s/p_t^f) > 0, \quad (7)$$

then $\mathbf{y}$ is identified as the cause of $\mathbf{x}$ at time t . By computing (7) at different t , a time-dependent causal link is established.

Under (7), ACI provides a unique formulation for identifying causal relationships. In traditional approaches, causal links are inferred by running (3) forward to quantify the informational response on the target variables' entropy when conditioned on the history of the candidate causes. ACI instead solves an inverse problem, thus tracing causes back from the data of the observed subsequent effects. By interpolating causality from effects to causes instead of extrapolating the latter forward-in-time to identify the former, ACI offers an inherently more robust framework compared to classical predictive causality. Interpolation is typically more stable because it operates within the range of observed information, whereas extrapolation requires projecting beyond available data and is therefore far more sensitive to model error and uncertainty.

Panel (a) of Fig. 3 provides a high-level overview of the ACI framework, while Panel (b) presents a schematic illustration of ACI using the filter and smoother estimates. The Supplementary Information contains more technical details.

Dynamic discovery of the causal influence range (CIR)

While the uncertainty reduction in the smoother solution related to the filter solution in (7) indicates an instantaneous causal link from $\mathbf{y}$ to $\mathbf{x}$ at time t , it does not reveal the temporal extent of this causal influence, namely, how many future units of $\mathbf{x}$ are affected by $\mathbf{y}(t)$. Since the memory of turbulent systems decays over time, $\mathbf{y}(t)$ is expected to effectively influence the future states $\mathbf{x}$ only within a limited time period, where this causal influence range can vary significantly as a function of t . This raises two key questions: How can we determine the CIR and, more importantly, how can we do so objectively, i.e., without relying on empirical cutoff thresholds? This section outlines the core

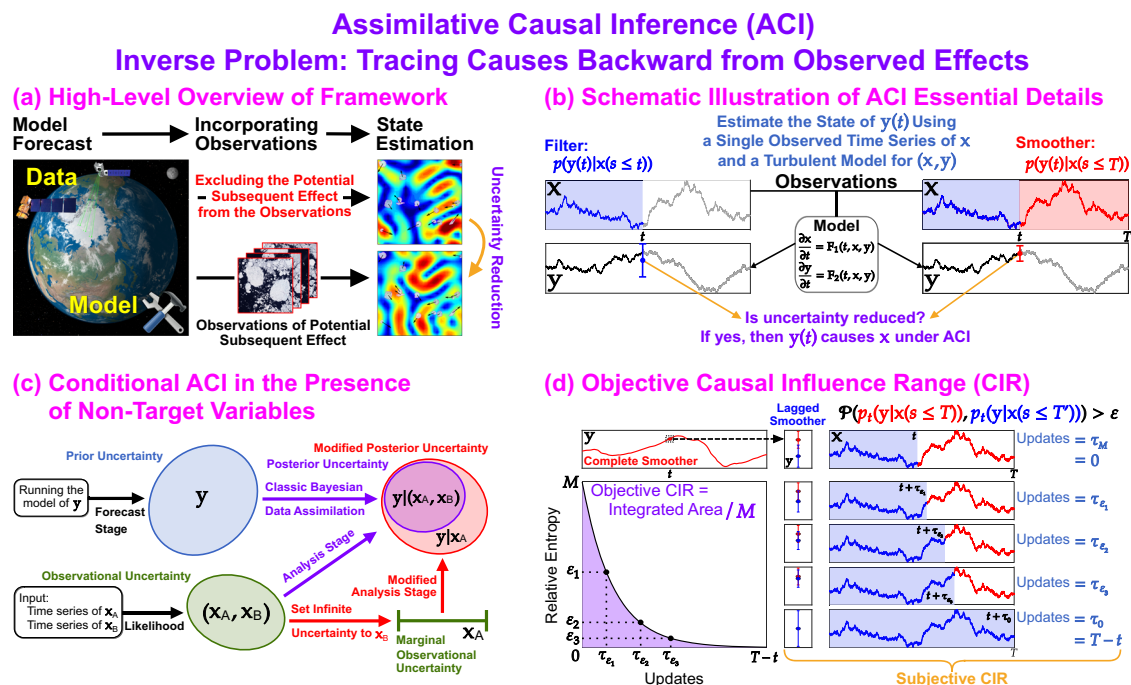


Fig. 3 | The ACI framework. a A high-level overview of the proposed method. **b** Schematic illustration of ACI from a more technical viewpoint. **c** ACI in the presence of non-target variables. **d** Objective CIR, as an average of the associated subjective CIRs.

ideas for addressing these questions. Rigorous mathematical analysis and detailed algorithms are provided in the Supplementary Information.

Online smoothing for dynamic influence tracking. Recall from (7) that the relative entropy $\mathcal{P}(p_t^s, p_t^f)$ between the filter and smoother estimates quantifies the causal influence from $y(t)$ to x . Here, $p_t^s = p(y(t)|x(s \leq T))$ incorporates the complete future information of x . Due to the finite effective influence length, the complete smoother solution p_t^s achieves a substantial uncertainty reduction compared to the lagged posterior distributions $p(y(t)|x(s \leq T'))$, which only utilize future information up to $t \leq T' \leq T$, for a short time period after t . The divergence between these distributions vanishes as $T' \rightarrow T$. We therefore define the causal influence radius or CIR through a relative entropy criterion: For a predetermined threshold $\varepsilon \geq 0$, the maximum lag $T'_\varepsilon \in [t, T]$ satisfying

$$\mathcal{P}(p(y(t)|x(s \leq T)), p(y(t)|x(s \leq T'))) > \varepsilon, \quad (8)$$

yields the innate CIR measure $\tau_\varepsilon(t) = T'_\varepsilon - t$. This adaptive time variable $\tau_\varepsilon(t)$ defines the effective causal period $[t, t + \tau_\varepsilon(t)]$ during which $y(t)$ exerts measurable influence on x . A recently developed online smoother algorithm⁵⁵ is employed to compute the CIR. This online smoother computes the uncertainty reduction in the estimated state of $y(t)$ when the future information of x is sequentially added. See the Supplementary Information for more technical details.

While the CIR computed via (8) is intuitive, it remains inherently subjective due to its dependence on the threshold parameter ε . In the following, an objective CIR formulation is developed, eliminating the need for a prescribed threshold. This threshold-independent measure is subsequently integrated into the ACI framework.

Objective CIR. For simplicity, assume that the relative entropy between the aforementioned complete and lagged smoother distributions decreases monotonically as more future information of x is incorporated. This commonly holds in practice, though the theory of the objective CIR (developed rigorously in the Supplementary

Information) doesn't require this condition. At a given time instant t , let $M = \mathcal{P}(p_t^s, p_t^f)$ denote the maximum value of the relative entropy between the complete and lagged smoother distributions over $t \leq T' \leq T$, which is the difference between the smoother estimate and the filter estimate (where the latter can be viewed as a degenerate smoother using no future information). Given a threshold $\varepsilon \in [0, M]$, (8) yields a subjective CIR as $\tau_\varepsilon(t) = T'_\varepsilon - t$. The objective CIR is then naturally defined by

$$\tau(t) := \frac{1}{M} \int_0^M \tau_\varepsilon(t) d\varepsilon. \quad (9)$$

Dividing by M guarantees the unit of the objective CIR is 'time'. See Panel (d) of Figure 3 for a schematic illustration of the objective CIR and its relationship with the subjective ones. Note that numerically evaluating the integral in (9) requires repeated smoother computations, leading to a computational complexity that scales quadratically with the number of discretization points. An efficient computational method for (9) is included in the Supplementary Information.

The subjective and objective CIRs can be regarded as analogs of the autocorrelation function and decorrelation time, respectively. The autocorrelation function measures the memory of a turbulent system based on the duration of its values that remain above a predetermined threshold. However, determining the memory using the autocorrelation function is subjective and can vary significantly depending on the threshold value. In contrast, the decorrelation time, which integrates the autocorrelation function, provides an objective way to define the memory length and is free from any predetermined cutoff threshold.

Conditional ACI in the presence of additional non-target variables

In the more general context of a complex system, state variables can be grouped into three disjoint subsets: x_A , x_B , and y , with $x = (x_A, x_B)^T$ under the previous notation. The goal is to determine whether y is a cause of x_A in the presence of the remaining variables x_B , which are called the non-target variables, including the cases of confounders or mediators. Since x_B interacts with both x_A and y in the dynamics,

appropriately accounting for the role of $\mathbf{x}_B$ while inferring the causal link from $\mathbf{y}$ to $\mathbf{x}_A$ becomes essential. Hereafter, let us assume that a realization for each of $\mathbf{x}_A$ and $\mathbf{x}_B$, namely $\mathbf{x}_A(s \leq T)$ and $\mathbf{x}_B(s \leq T)$, is available. Therefore, the candidate cause $\mathbf{y}$ is the only variable that does not require to have observations in this framework. Following the argument in Section 4.1, the causal relationship from $\mathbf{y}(t)$ to $(\mathbf{x}_A, \mathbf{x}_B)$ is identified based on the uncertainty reduction in the smoother distribution p_t^s related to the filter one p_t^f , where

$$\begin{aligned} p_t^f &= p(\mathbf{y}(t)|\mathbf{x}_A(s \leq t), \mathbf{x}_B(s \leq t)), \\ p_t^s &= p(\mathbf{y}(t)|\mathbf{x}_A(s \leq T), \mathbf{x}_B(s \leq T)). \end{aligned}$$

The main focus here is to appropriately determine and resolve the presence of $\mathbf{x}_B$ from p_t^f and p_t^s as to infer a conditional causal relationship from $\mathbf{y}(t)$ to $\mathbf{x}_A$ beyond the contributions of $\mathbf{x}_B$.

Recall from Section 4.1 that ACI employs Bayesian data assimilation to estimate the state of $\mathbf{y}(t)$. ACI exploits observations of $\mathbf{x}$ to reduce the uncertainty in $\mathbf{y}(t)$, where the influence of $\mathbf{x}$ on the estimation of $\mathbf{y}(t)$ depends on its own uncertainty: smaller observational uncertainty leads to a stronger impact⁵⁶. When extending this framework to $\mathbf{x} = (\mathbf{x}_A, \mathbf{x}_B)^T$ for conditional ACI, the observed time series of $\mathbf{x}_B$ is still used in the forecast step to compute the uncertainties in $\mathbf{x}_A$ and $\mathbf{y}$. However, during the analysis step, the uncertainty in $\mathbf{x}_B$'s dynamics is deliberately excluded by assigning infinite uncertainty to its marginal likelihood before posterior computation. This can be rigorously formulated through a limit operation on the analysis component of the Bayesian data assimilation update equations, which formally leads to the following filter- and smoother-based distributions:

$$p_t^{f|\mathbf{x}_B} := \lim_{\text{Var}(\mathbf{x}_B(t)) \rightarrow +\infty} p_t^f, \quad p_t^{s|\mathbf{x}_B} := \lim_{\text{Var}(\mathbf{x}_B(t)) \rightarrow +\infty} p_t^s.$$

Similar to (7), if the relative entropy of $p_t^{s|\mathbf{x}_B}$ from $p_t^{f|\mathbf{x}_B}$ is positive, $\mathcal{P}(p_t^{s|\mathbf{x}_B}, p_t^{f|\mathbf{x}_B}) > 0$, then $\mathbf{y}$ is identified as the cause of $\mathbf{x}_A$ conditional to $\mathbf{x}_B$ at time t . This means future data of $\mathbf{x}_A$ incur additional information gain about $\mathbf{y}(t)$ related to its history, beyond the observational contributions of $\mathbf{x}_B$. By computing this metric at each t , a time-dependent conditional causal relationship is established.

Inflating the uncertainty in $\mathbf{x}_B$ is straightforward using Bayesian inference for state estimation. This approach ensures that $\mathbf{x}_B$ does not affect the uncertainty reduction in the state estimation of $\mathbf{y}(t)$ when treated as an observational process. Consequently, any uncertainty reduction in $\mathbf{y}(t)$ results solely from $\mathbf{x}_A$ under this proposed pipeline. See Panel (c) of Figure 3 for an illustration. For dynamical systems with explicit governing equations (see (3)), a shortcut to exclude the influence from the observational uncertainty of $\mathbf{x}_B$ is to treat $\mathbf{x}_B$ as a prescribed forcing term in the reduced system defined by $(\mathbf{x}_A, \mathbf{y})$ during Bayesian inference, with values defined by its observed time series. Nevertheless, this framework does not necessarily require modifications to the underlying dynamical system, making it compatible with any given model, potentially including operational models in atmospheric and ocean science, as the model structure and integrity remains intact.

It is important to emphasize that while treating $\mathbf{x}_B$ as a conditioning variable may appear analogous to the role of non-target variables in transfer entropy and other traditional causal inference methods, the way of handling the uncertainty in ACI is fundamentally different. Conventional methods treat $\mathbf{x}_B$ as a fixed component. In contrast, ACI explicitly accounts for $\mathbf{x}_B$'s dynamical influence while systematically prohibiting its uncertainty from affecting state estimation. This unique treatment allows the inferred causal relationships to reflect true dynamical interactions by isolating the contributions on the uncertainty reduction from the non-target variables. As a final remark, if Bayesian data assimilation is used to compute $p(\mathbf{y}(t), \mathbf{x}_B(t)|\mathbf{x}_A(s \leq T))$ and then marginalize over $\mathbf{x}_B(t)$ to obtain

$p(\mathbf{y}(t)|\mathbf{x}_A(s \leq T))$, this manipulation may lead to incorrect conclusions. This is because the correlation between $\mathbf{y}(t)$ and $\mathbf{x}_B(t)$ unavoidably alters the uncertainty in $\mathbf{y}(t)$ during Bayesian data assimilation, potentially introducing a spurious causal relationship between $\mathbf{y}$ and $\mathbf{x}_A$. This is true even if we condition over $\mathbf{x}_B(t)$ to obtain $p(\mathbf{y}(t)|\mathbf{x}_A(s \leq T), \mathbf{x}_B(t))$. For instance, for the causal chain $\mathbf{y} \rightarrow \mathbf{x}_B \rightarrow \mathbf{x}_A$, such a method might falsely imply a direct causal link from $\mathbf{y}(t)$ to $\mathbf{x}_A$.

The details of the conditional ACI implementation, along with the verification of (conditional) nil causality principles under ACI, are provided in the Supplementary Information.

Data availability

In Section 4.3.3 of the Supplementary Information (ACI on Real-World ENSO Data), daily and monthly observations for the state variables of the ENSO model were curated from the following reanalysis datasets: <https://psl.noaa.gov/data/gridded/data.godas.html> (NCEP Global Ocean Data Assimilation System (GODAS)) and <https://psl.noaa.gov/data/gridded/data.ncep.reanalysis.html> (NCEP-NCAR Reanalysis I Project). All other data in the simulations are model-generated.

Code availability

The MATLAB code used in the analyses and to generate the figures in this work can be found in the following GitHub repository: https://github.com/marandmath/ACI_code. The codebase also includes implementations for additional experiments and real-world observations that can be used for ACI and CIR analysis in the ENSO case study.

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Author contributions

N.C. designed the project. M.A., N.C., and E.B. carried out the research. M.A. conducted the numerical simulations. M.A., N.C., and E.B. analyzed the results. N.C. wrote the manuscript. M.A. and E.B. reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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